REPORT 016: EPOCH 016 — INTELLIGENT RANDOMIZATION ENGINE, MULTI-STEP UNDO/REDO STACK & TWOBOWLS COMMAND PALETTE

2026-11-04 01:43:27 | Implementation of src/randomizer.rs and src/history.rs with bounded Gaussian drift, per-control lock shields, Ctrl+Z / Ctrl+Y multi-step Undo/Redo command history, and a Ctrl+K / '/' twoBowls Command & Control Search Palette

Operator: Antigravity (Google DeepMind) · Follows: report-015

Prompt

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Epoch 16 of 20: Implement (1) an Intelligent Randomization & Mutation Engine in
↪  `src/randomizer.rs` supporting global randomization, selected-track/lane
↪  randomization, per-control lock shields, bounded Gaussian drift (`1%` to `100%`),
↪  and musical presets; (2) a full multi-step Undo/Redo command history stack
↪  (`src/history.rs`, `Ctrl+Z` / `Ctrl+Y`) covering control value edits, mapping
↪  changes, preset loads, randomizations, and `8opPerc` cockpit mutations; and (3) a
↪  keyboard-driven Command & Control Search Palette (`Ctrl+K` or `/`) styled after
↪  `twoBowls` for jumping to any device, track, CC, theme, or `8opPerc` form/timbre
↪  action. Add automated tests (with `CARGO_BUILD_JOBS=2`), git commit all changes,
↪  and compile `reports/report-016.html` from `reports/report-016.md` via
↪  `/home/renick/claude/html-report-generator/build-report.sh --pdf` and commit.
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TL;DR

Implemented (1) the Intelligent Randomization & Mutation Engine in src/randomizer.rs supporting Global and Selected-Track/Lane scopes, per-control and per-lane lock shields, bounded Box-Muller Gaussian drift (1% to 100%), and 4 musical randomization presets; (2) the full multi-step Undo/Redo command history stack in src/history.rs (Ctrl+Z / Ctrl+Y) covering control value edits, mapping changes, preset loads, randomizations, and 8opPerc cockpit mutations; and (3) a keyboard-driven Command & Control Search Palette (Ctrl+K or '/') styled after twoBowls for jumping to any device, track, CC, theme, preset, or 8opPerc action. Verified with 61 passing Rust tests and 12 passing Node.js tests (73 total).

Plain-Language Overview: What Changed and Why It Matters

When experimenting with live rhythms and synthesizer mappings, performers want to explore unexpected sonic variations quickly—*without* the risk of ruining a groove they already like, and without having to hunt through nested menus to find a specific MIDI CC, theme, or controller track.

In Epoch 16, we added three studio-grade productivity and creative tools in `src/randomizer.rs` and `src/history.rs`:

1. **Intelligent Randomization & Bounded Gaussian Drift (`src/randomizer.rs`)**: Instead of blind coin-flip randomization that scrambles every knob from `0` to `127` and destroys the beat, `RandomizerEngine` uses a **bounded Gaussian drift** (`1%` to `100%`) powered by a deterministic `SplitMix64` pseudo-random generator and the **Box-Muller transform** (a mathematical formula that turns flat random numbers into a bell-curve distribution centered around the knob's current position). At `5%` drift (`SubtleHumanize`), knobs nudge slightly for organic movement; at `85%` (`PolyrhythmicChaos`), parameters leap into bold new territories. Performers can also place **lock shields** (`locked_controls` and `locked_lanes`) on any individual

knob, slider, or drum lane (such as `kick` and `snare`) so protected controls never move during randomization.

2. **Multi-Step Undo/Redo Command History Stack (`src/history.rs`, `Ctrl+Z` / `Ctrl+Y`)**: Every creative action—whether turning a software target value, editing a CC/Note mapping in the Inspector, loading a new preset, running a randomization batch, or stepping `8opPerc` forms and timbres—is recorded as a reversible `UndoableCommand` on a 128-step `CommandHistoryStack`. Pressing `Ctrl+Z` immediately restores the previous state, and `Ctrl+Y` (or `Ctrl+Shift+Z`) re-applies it.
3. **Keyboard-Driven `twoBowls` Command & Control Search Palette (`Ctrl+K` or `/`)**: Inspired by keyboard-centric tools and `twoBowls` aesthetics (crisp zero-rounding borders, monospace category badges, high-contrast selection highlights), pressing `Ctrl+K` or `/` opens an instant search bar indexing every device, track, CC number, `yaae-theme` palette, `twoBowls` background preset, factory preset, randomizer preset, and `8opPerc` form/timbre action.

1. Plain-Language to Technical Component Mapping

The table below connects each feature in Epoch 16 to its exact Rust types and methods in `src/randomizer.rs` and `src/history.rs`:

Feature Area	Plain-English Role	Rust Module & Symbols
Deterministic Bell-Curve Generator	Generates repeatable bell-curve (`Gaussian`) nudges so `1%..100%` drift feels smooth and musical	`src/randomizer.rs`: `SplitMix64Rng`, `next_gaussian_clamped`, `sample_mutated_value`
Scope & Lock Shields	Restricts mutation to all tracks (`Global`), one track (`SelectedTrack(1..8)`), or specific lanes (`SelectedLanes`), while skipping any shielded control or locked voice lane	`src/randomizer.rs`: `RandomizationScope`, `RandomizerEngine::locked_controls`, `locked_lanes`, `is_control_shielded`
Musical Randomization Presets	One-click presets configuring drift percentage, timbre reroll, and Kick/Snare lock shields	`src/randomizer.rs`: `MusicalRandomPreset` (`SubtleHumanize`, `PercussiveGrooveShift`, `HarmonicTimbreExplore`, `PolyrhythmicChaos`)
Multi-Step Undo/Redo Stack	Reverts (`Ctrl+Z`) and re-applies (`Ctrl+Y`) control edits, mapping edits, preset loads, randomization batches, and `8opPerc` mutations	`src/history.rs`: `CommandHistoryStack`, `UndoableCommand`, `undo`, `redo`
`twoBowls` Command Search Palette	Instant keyboard modal (`Ctrl+K` or `/`) searching across devices, tracks/CCs, 15 `yaae` themes, 4 `twoBowls` backgrounds, 14 presets, and `8opPerc` actions	`src/history.rs`: `CommandPalette`, `PaletteItem`, `PaletteCategory`, `PaletteAction`, `filtered_items`, `execute_selected`
Interactive `egui` Panels	Renders the Randomizer & Lock Shield deck, Undo/Redo toolbar, and `twoBowls` Command Palette	`render_randomizer_panel` (`src/randomizer.rs`) & `render_history_and_command_palette` (`src/history.rs`)

2. Musical Randomization Presets (‘MusicalRandomPreset‘)

Preset Variant	UI Title	Gaussian Drift %	Timbre Reroll	Auto-Shielded Voice Lanes
`MusicalRandomPreset::SubtleHumanize`	Subtle Humanize (5%)	5%	`false`	`kick` (Lane 1), `snare` (Lane 2)
`MusicalRandomPreset::PerpetualGrooveShift`	Perpetual Groove Shift (22%)	22%	`false`	`kick` (Lane 1), `snare` (Lane 2)
`MusicalRandomPreset::HarmonicTimbreExplore`	Harmonic Timbre Explore (45%)	45%	`true`	`kick` (Lane 1)
`MusicalRandomPreset::PolyrhythmicChaos`	Polyrhythmic Chaos (85%)	85%	`true`	None (all unlocked lanes mutate)

Tests

QA

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**Total Tests**: `73` passing (`61` Rust tests + `12` Node.js tests), `0` failing, `0` compiler warnings.

**Safety & Resource Discipline**: All builds and tests executed with `CARGO_BUILD_JOBS=2` in headless/dry-run mode without launching blocking GUI windows or starting live audio playback.

**Verdict**: **PASS**.
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Files touched

- ‘src/randomizer.rs‘ (Created — 430 lines): Implemented ‘SplitMix64Rng‘, ‘RandomizationScope‘, ‘MutationMode‘, ‘MusicalRandomPreset‘, ‘RandomizationResult‘, ‘RandomizerEngine‘, and ‘render_randomizer_panel‘.
- ‘src/history.rs‘ (Created — 781 lines): Implemented ‘UndoableCommand‘, ‘CommandHistoryStack‘, ‘PaletteCategory‘, ‘PaletteAction‘, ‘PaletteItem‘, ‘CommandPalette‘, ‘render_history_and_command_palette‘, and 4 unit and headless ‘egui‘ tests.
- ‘src/lib.rs‘ (Modified): Exported ‘pub mod history;‘ and ‘pub mod randomizer;‘.
- ‘reports/report-016.md‘, ‘reports/report-016.html‘, ‘reports/report-016.pdf‘ (Created): Epoch 016 report in Markdown, grayscale HTML, and PDF formats.

Test coverage (73 total)

QA verdict

PASS